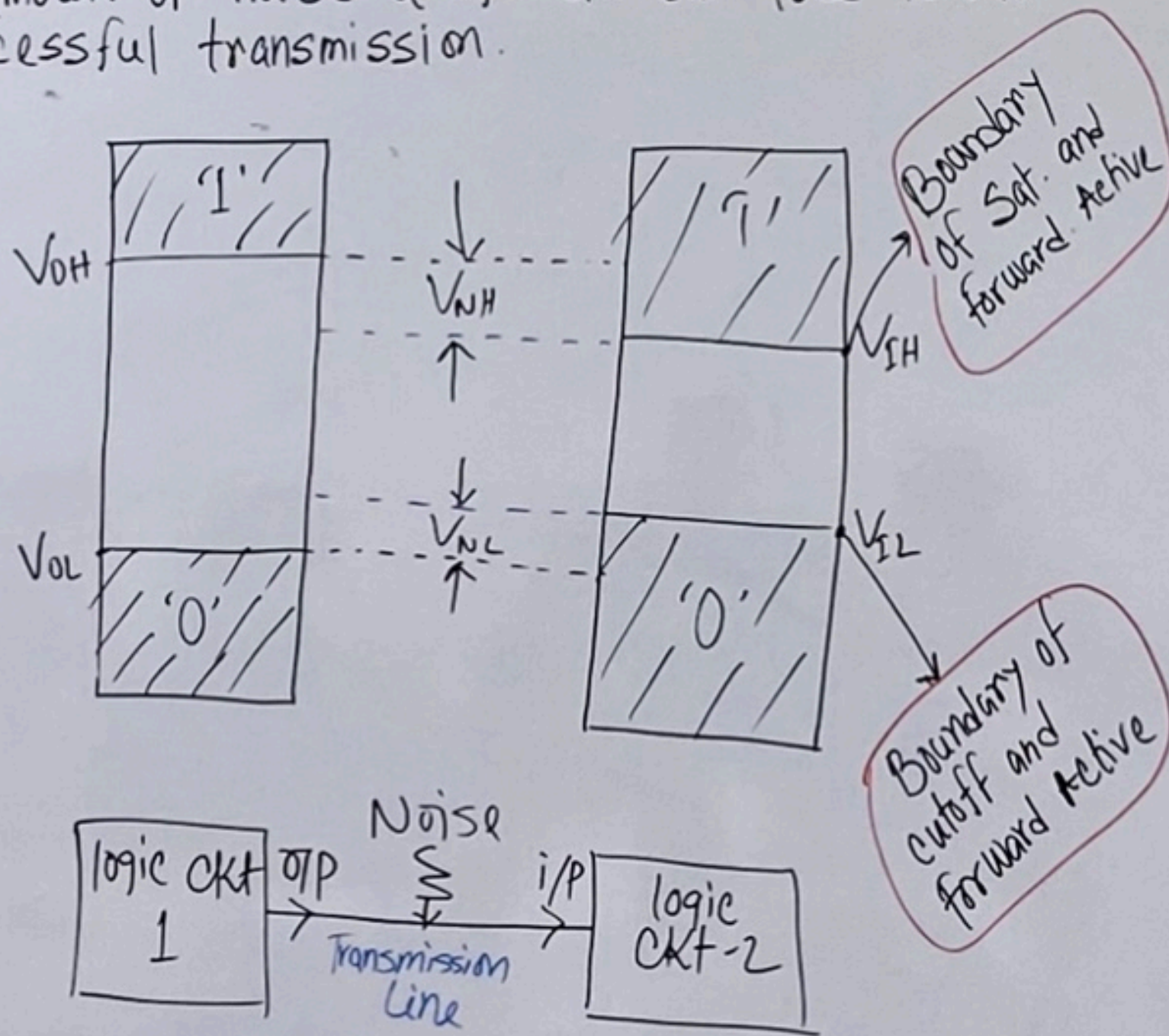


Noise Margin

The amount of noise a logic ckt can tolerate while successful transmission.

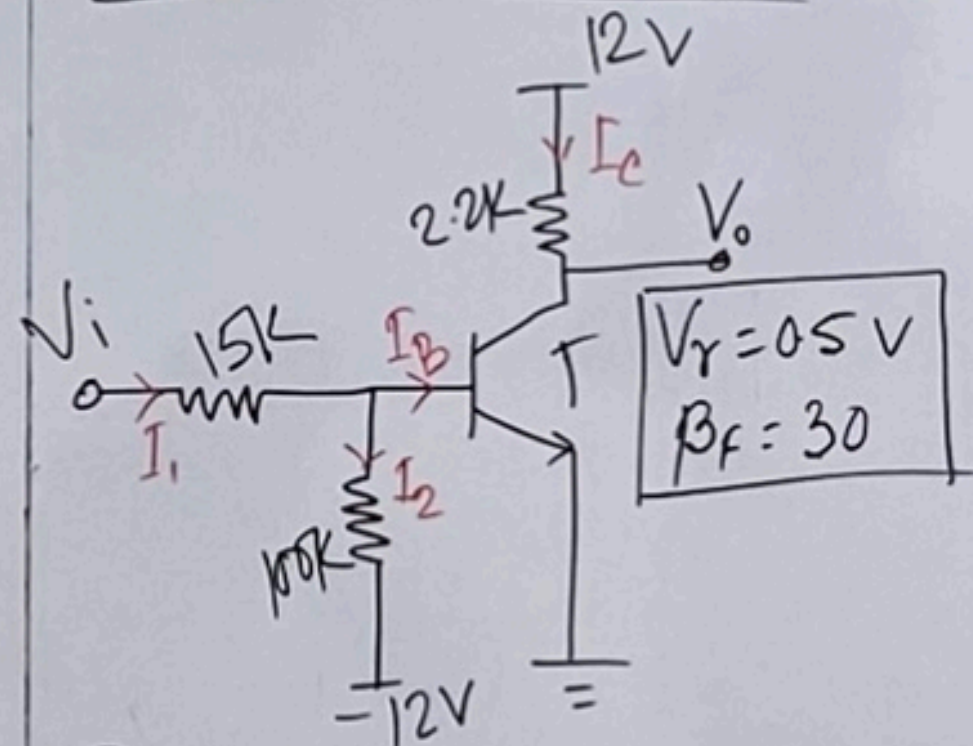


$$V_{NH} \rightarrow \text{high state Noise Margin} = V_{OH} - V_{IH}$$

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$$\text{Max noise Margin, } NM = \min(V_{NH}, V_{NL})$$

RTL Noise Margin



(iv) Find the Noise Margin of the above RTL ckt

$$V_{OL} = 0.2V, V_{OH} = 12V$$

V_{IL}

T → is and

But

V_{IL} = V_{IL}

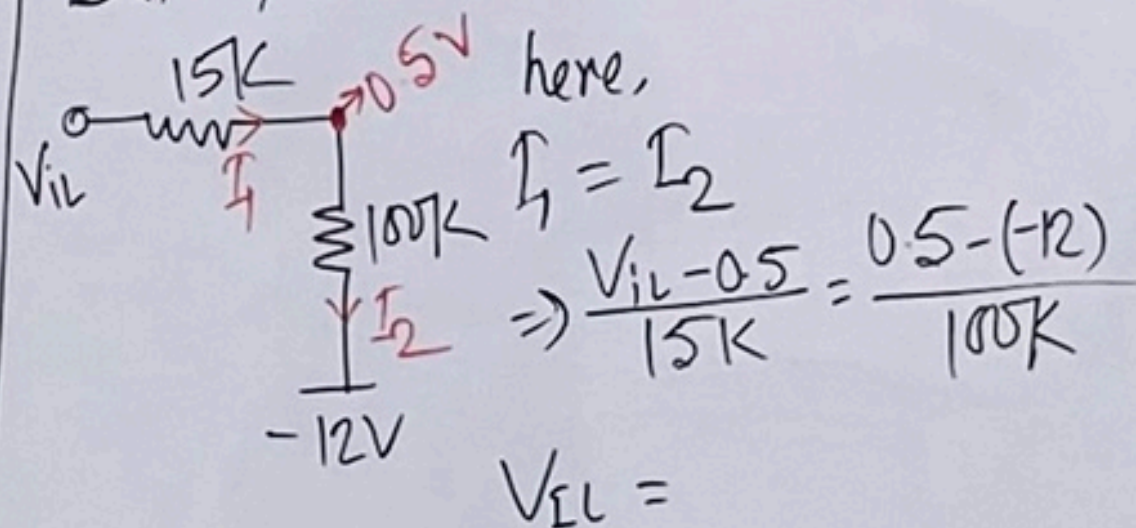
V_{NL} = V_{OL} - V_{IL}

$$\begin{matrix} V_{OH} - V_{IH} \\ V_{IL} - V_{OL} \end{matrix}$$

V_{IL} calculation

$T \rightarrow$ is on the boundary of Cutoff and forward Active $\rightarrow \therefore V_{BE} = V_T = 0.5V$
 $V_B = 0.5V$

But $T \rightarrow$ cutoff



$$\begin{aligned} V_{NL} &= V_{IL} - V_{OL} \\ &= -0.2 \end{aligned}$$

V_{IH} calculation

T is on the boundary of Sat. and F.A

$$\therefore \beta_{forced} = \beta_F = 30$$

But $T \rightarrow$ Sat. $\rightarrow V_{CE} = 0.2V, V_{BE} = 0.8V$
 in this case $\rightarrow V_O = 0.2V$

$$\therefore I_C = \frac{12 - 0.2}{2.2K}$$

$$\beta = \frac{I_C}{I_B}$$

$$\Rightarrow 30 = \frac{I_C}{I_B}$$

$$I_2 = \frac{0.8 - (-12)}{100K} = 0.128mA \Rightarrow I_B =$$

$$\therefore I_1 = I_2 + I_B =$$

$$I_1 = \frac{V_{IH} - 0.8}{15K}$$

$$\Rightarrow V_{IH} =$$

$$\therefore V_{NH} = V_{OH} - V_{IH} = 12 -$$

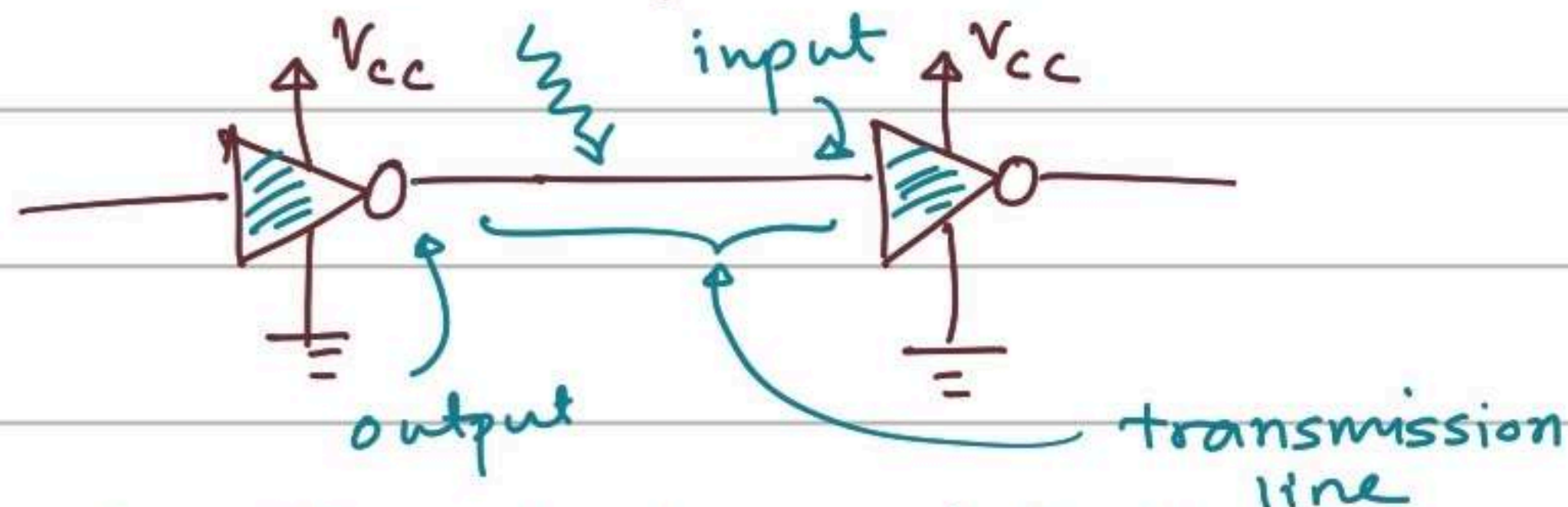
$$\therefore \text{Overall / Max Noise margin, } NM = \min(V_{NH}, V_{NL})$$

$$P_3 = (12 - 0) \times 5.3636 = 64.3632 \text{ mW}$$

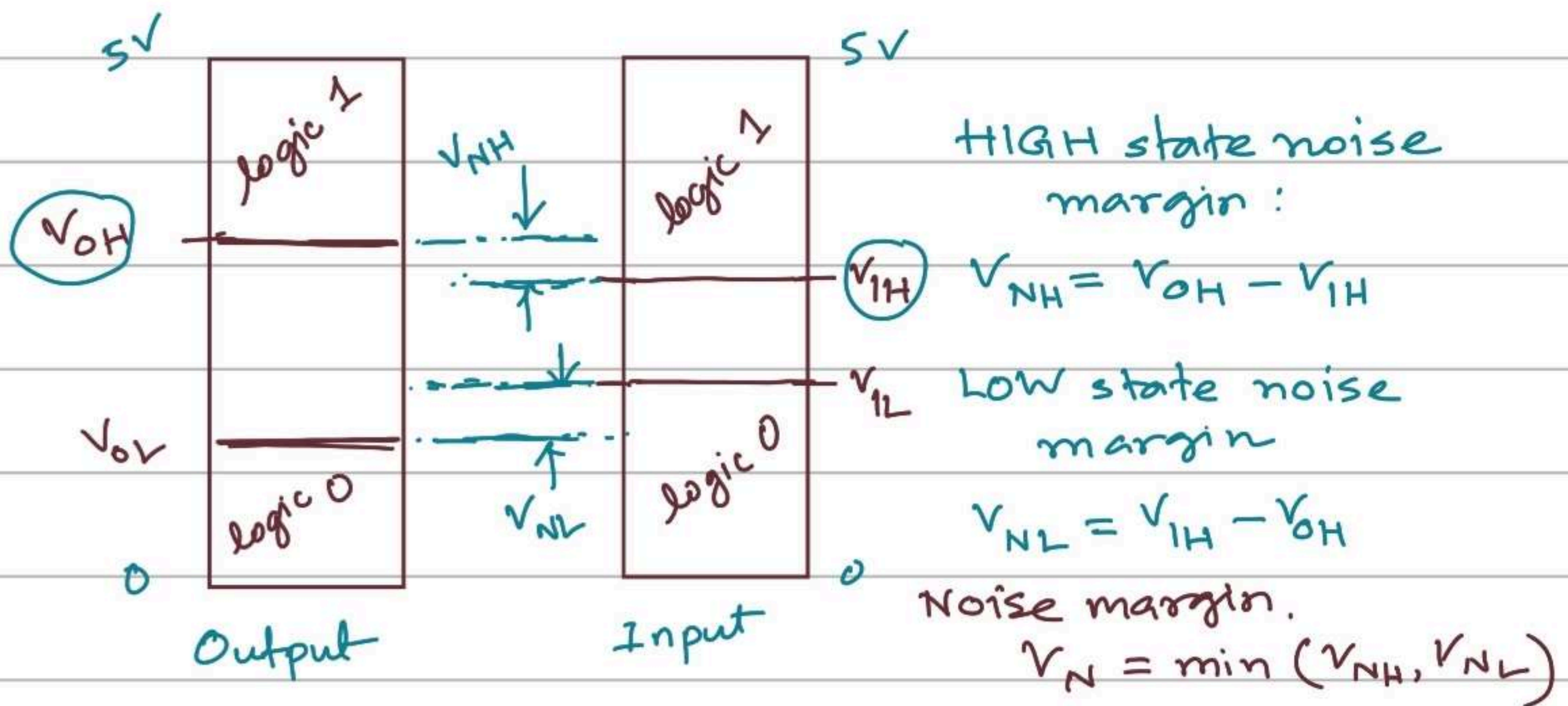
$$\text{Total power dissipation, } P = P_1 + P_2 + P_3 = \boxed{74.8534 \text{ mW}}$$

NOISE MARGIN:-

The amount of noise voltage a ^{digital logic} circuit can tolerate while successful transmission.



We model different types interference or noises using "noise voltage."



Why do we use minimum?

→ To ensure safety margins for both high and low voltages.

$\frac{V_{OH}}{V_{IH}}$ → minimum voltage level at an output in the logical "1" state under defined load condition.

$V_{OL} \rightarrow$ maximum " " at an output in the logical "0" state under defined ^{input} load condition.

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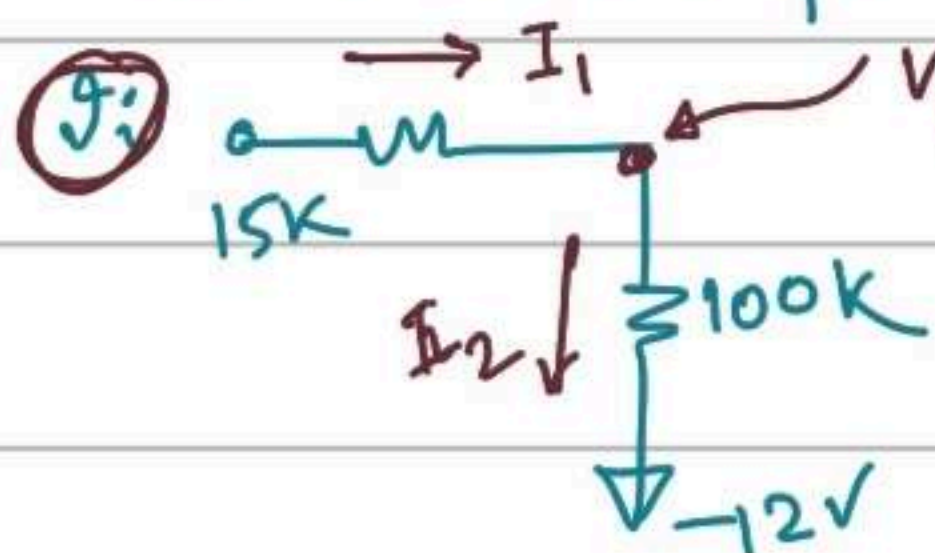
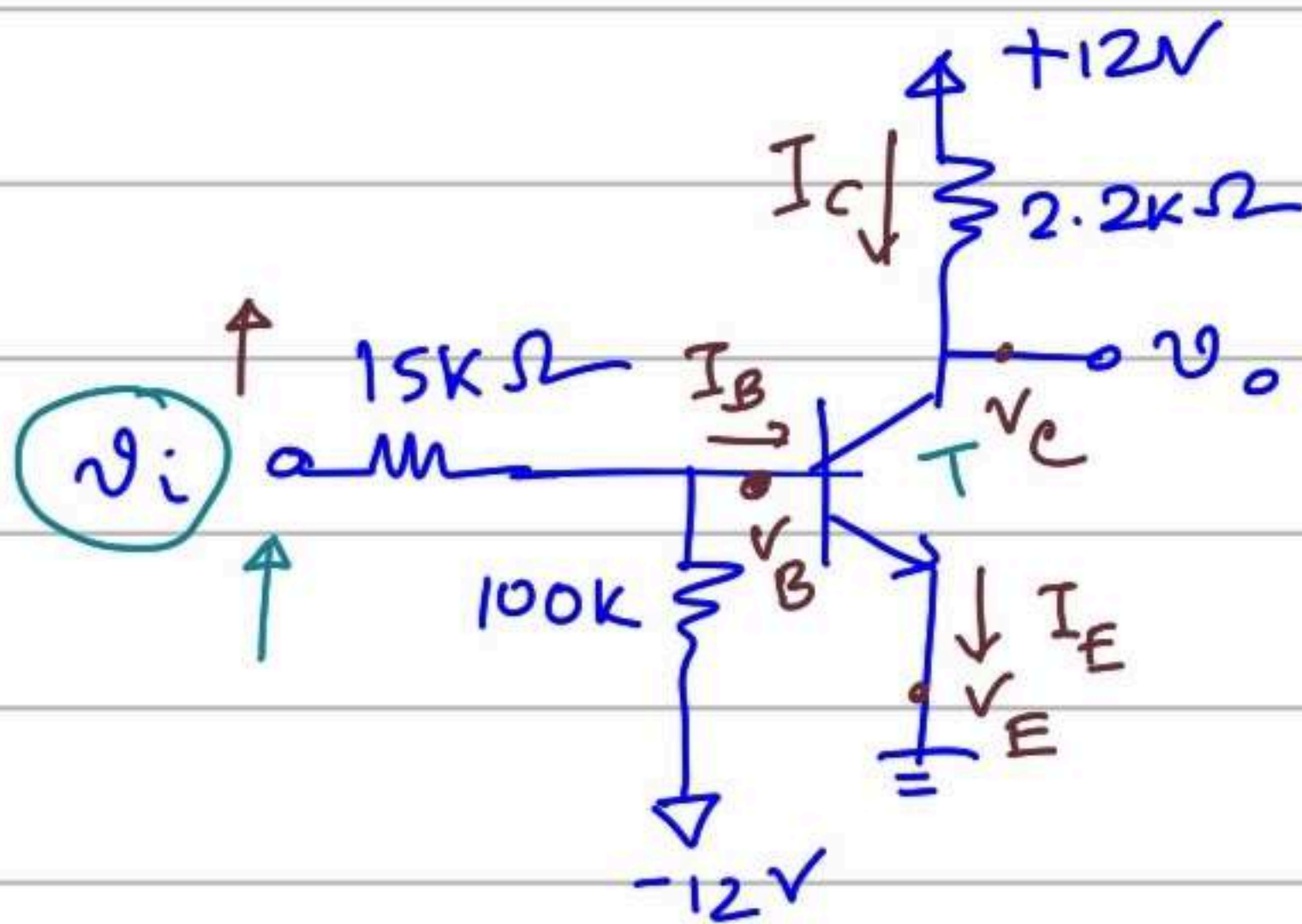
$V_{BE} = 0.5V$ but we assume transistor is in cutoff.

* T $\rightarrow$ cutoff, $I_B = 0mA$ $I_2 = \frac{0.5 - (-12)}{100k} = 0.125mA$

$I_1 = I_2$, $I_1 = \frac{V_i - 0.5}{15k} = 0.125mA$

$\therefore V_i = 0.5 + 15 \times 0.125 = 2.375V$

$V_{IL} = 2.375V$



V_{IH} calculation: When input voltage is high, transistor

T operates in saturation mode. $V_{CE} = 0.2V = v_o$.

* The minimum voltage at the input terminal which will change the operating mode of transistor

T to saturation to forward active mode should be V_{IH} .

* We will assume T is saturation,

but $\beta_{forced} \approx \beta_F$. [Transition point between sat. and F.A.]

* T $\rightarrow$ saturation. $I_C = \frac{12 - 0.2}{2.2k} = 5.3636 \text{ mA}$

$$\beta_{forced} = \frac{I_C}{I_B} = 30 \Rightarrow I_B = \frac{I_C}{\beta} = 0.1788 \text{ mA}.$$

$$I_2 = \frac{0.8 - (-12)}{100k} = 0.128 \text{ mA}. \quad I_1 = I_2 + I_B = 0.3068 \text{ mA}$$

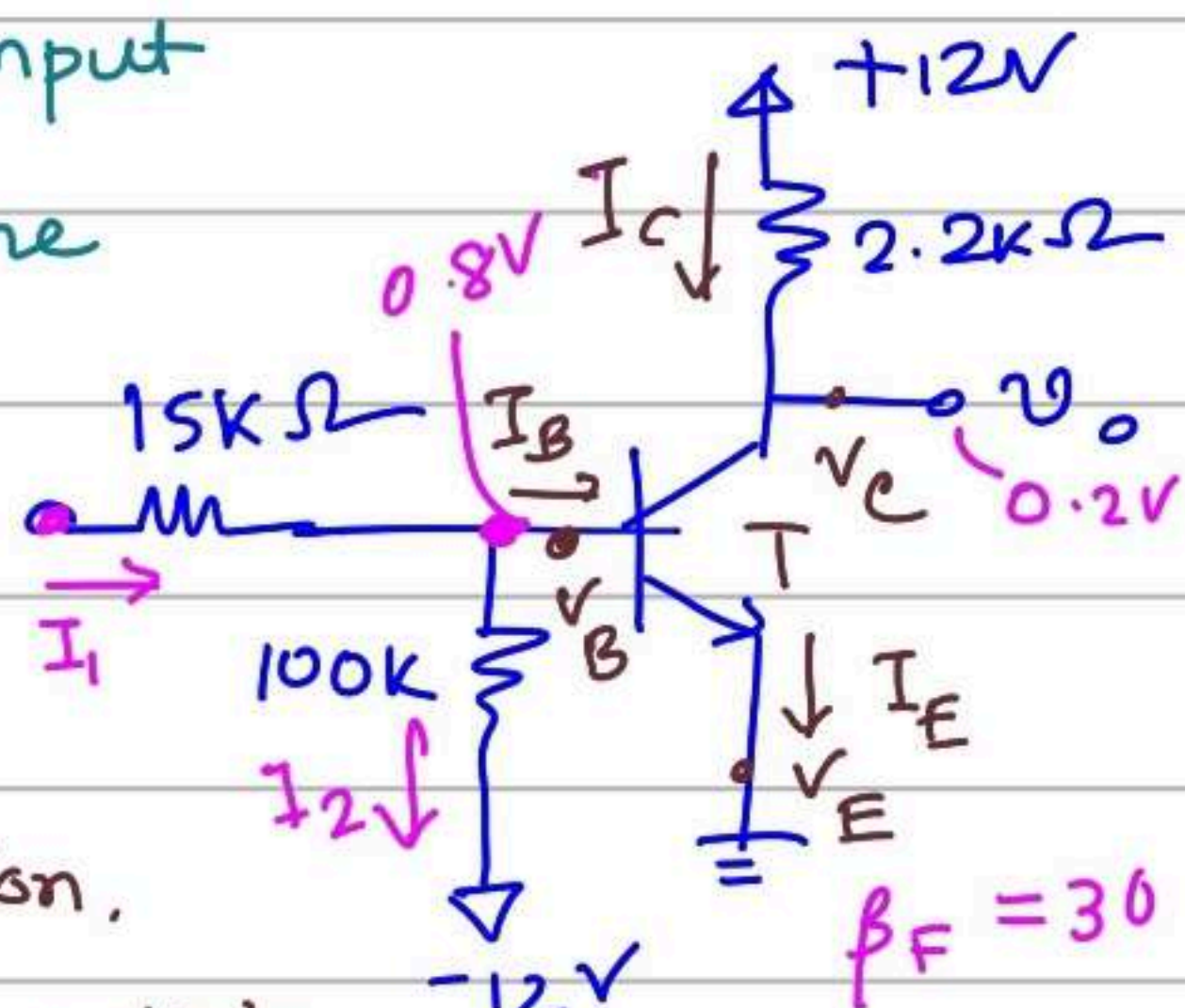
$$\frac{v_i - 0.8}{15k} = 0.3068 \text{ mA} \Rightarrow v_i = 0.8 + 15 \times 0.3068 = 5.4018 \text{ V}$$

$$\boxed{V_{IH} = 5.4018 \text{ V}}$$

$$\text{High state noise margin: } V_{NH} = V_{OH} - V_{IH} = 6.0982 \text{ V}$$

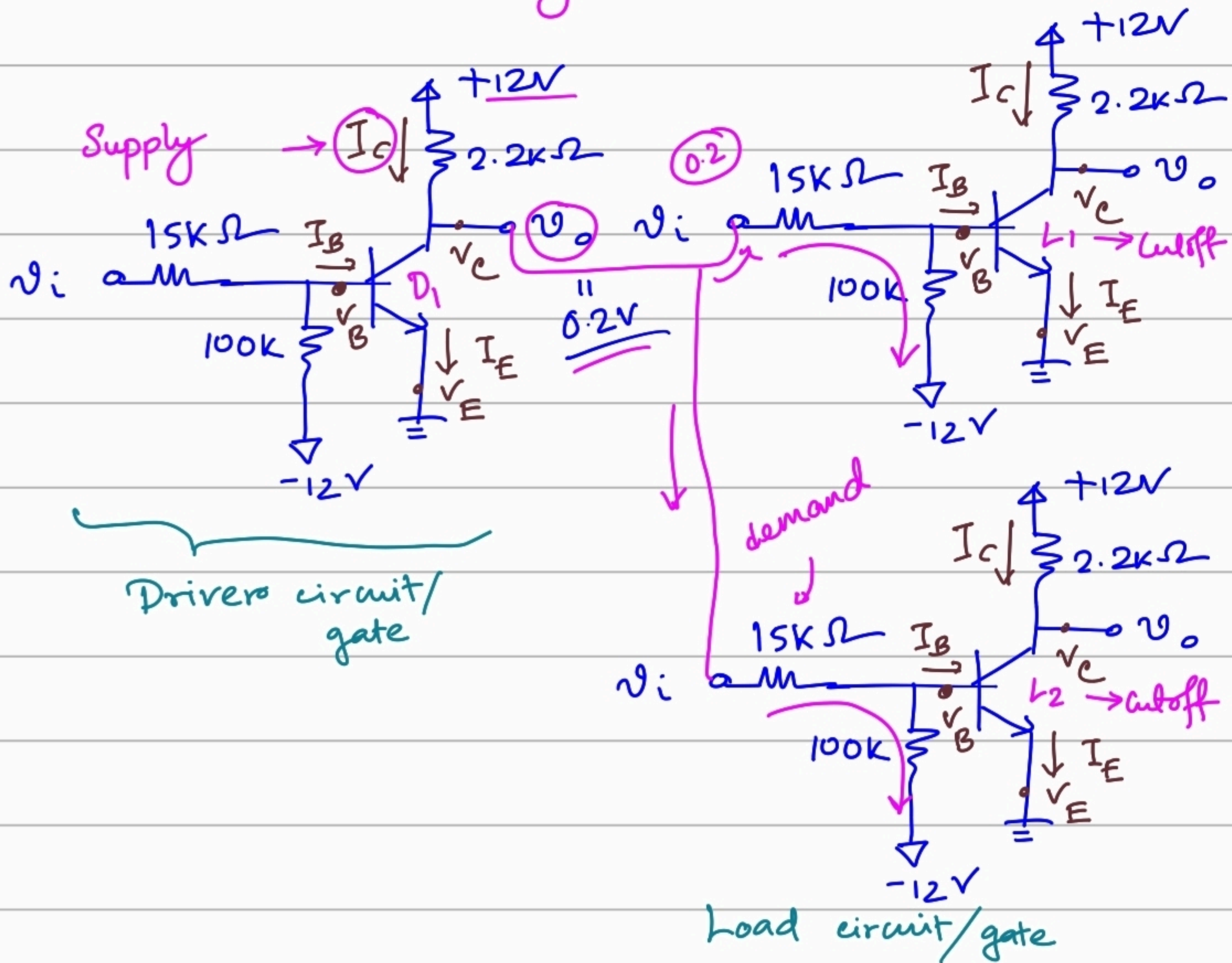
$$\text{Low state noise margin: } V_{NL} = V_{IL} - V_{OL} = 2.175 \text{ V}$$

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Fanout: The number of input terminals are connected to a output terminal.

Maximum fanout (fanout): The maximum number of logic inputs of the logic family that an output terminal can drive, is called maximum fanout.
reliably

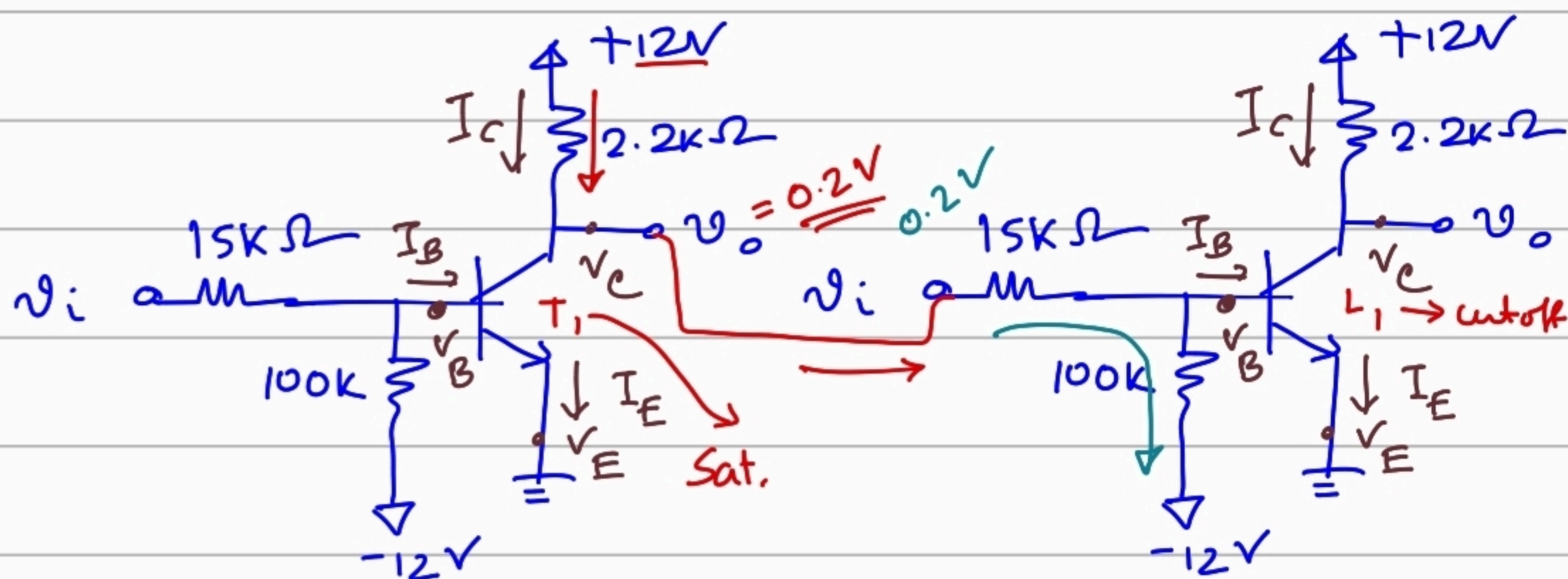


Question: Calculate the maximum fanout of the above RTL circuit. Assume $V_{OH} = 10V$.

Case ①: The output of the drivers circuit is low.

* Maximum supply current :- The maximum of current can be delivered by the driver circuit without changing its operating mode.

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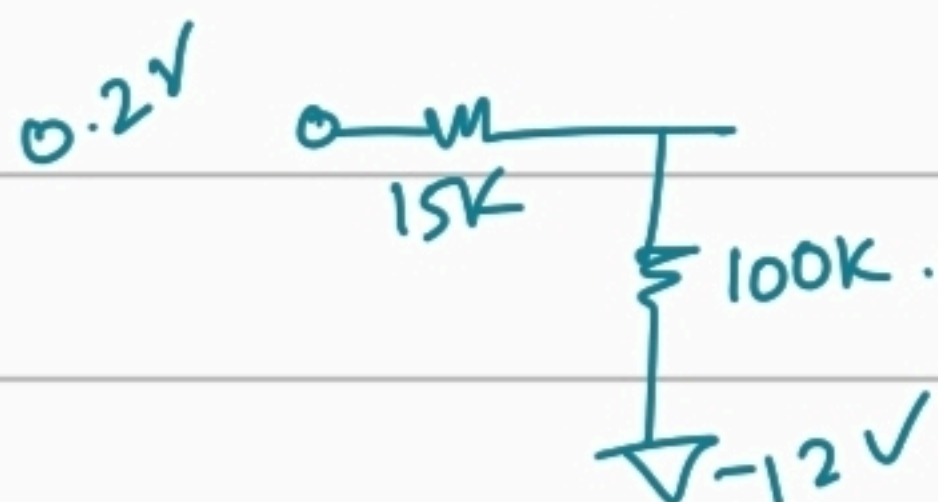


Driver circuit $T_1 \rightarrow$ Saturation. $V_o = 0.2V$

Load circuits $L_1, L_2 \rightarrow$ cutoff. $I_B = 0$.

* Maximum supply current, $\frac{12-0.2}{2.2k} = 5.3636 \text{ mA}$

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maximum fanout for this case,

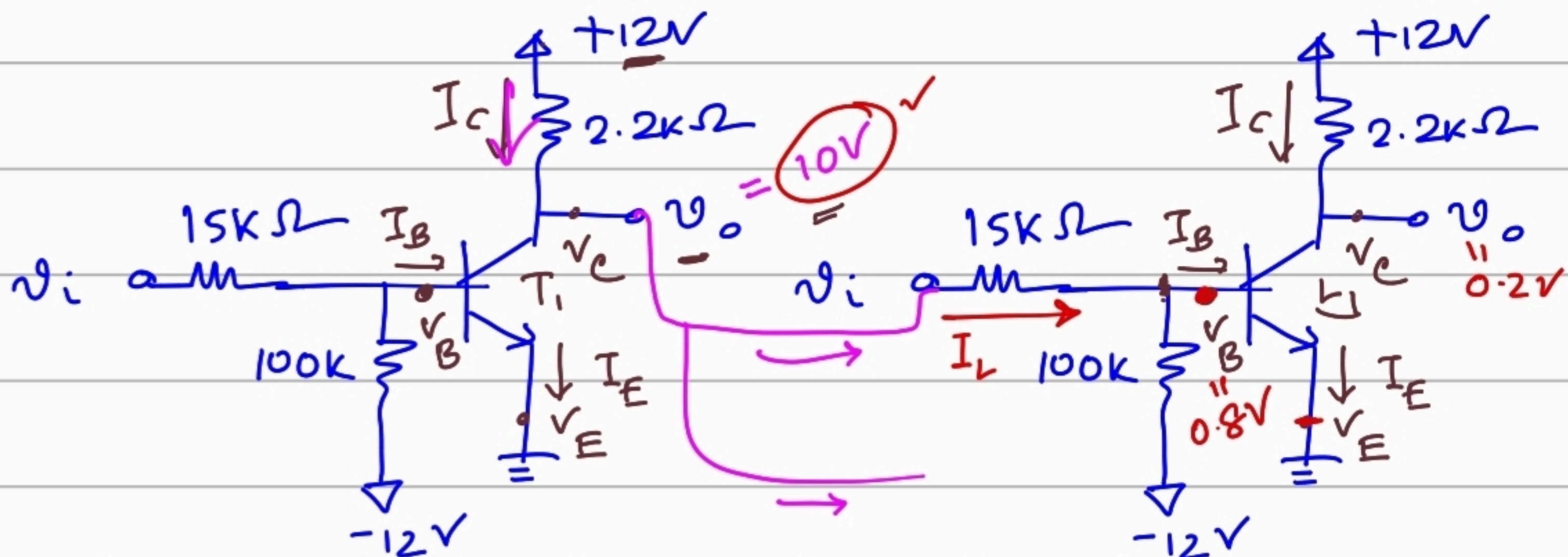
$$= \left\lfloor \frac{5.3636}{0.106} \right\rfloor = \left\lfloor 50.6 \right\rfloor = 50$$

* If we connect 51 load circuits, then demand exceeds supply. Thus driver circuit will malfunction.

Case ② : The output voltage of the driver circuit is high.

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$$V_{OH} = 10V$$



* Driver transistor, $T_1 \rightarrow$ cutoff, load circuit's transistors T_2

* Maximum supply current = $\frac{12-10}{2.2k\Omega} = 0.909mA$ Sat.

* Individual load current, $I_L = \frac{10-0.8}{15k} = 0.6133mA$

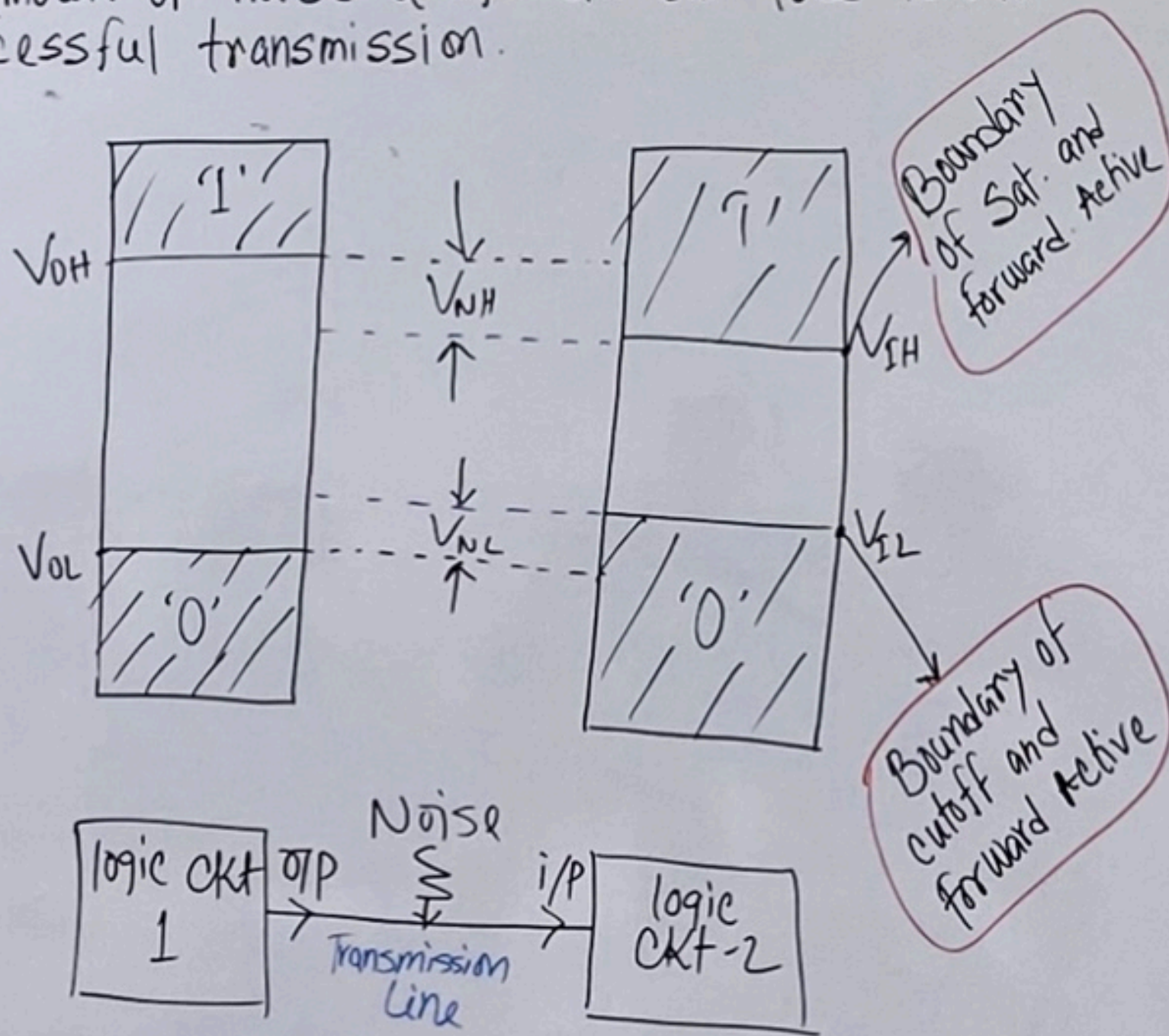
* Maximum fanout for this case = $\left\lfloor \frac{0.909}{0.6133} \right\rfloor = \lfloor 1.48 \rfloor = 1$

□ Overall maximum fanout must be calculated by considering worst case scenario. To prevent malfunction of driver circuit, we need to take the minimum of the two cases.

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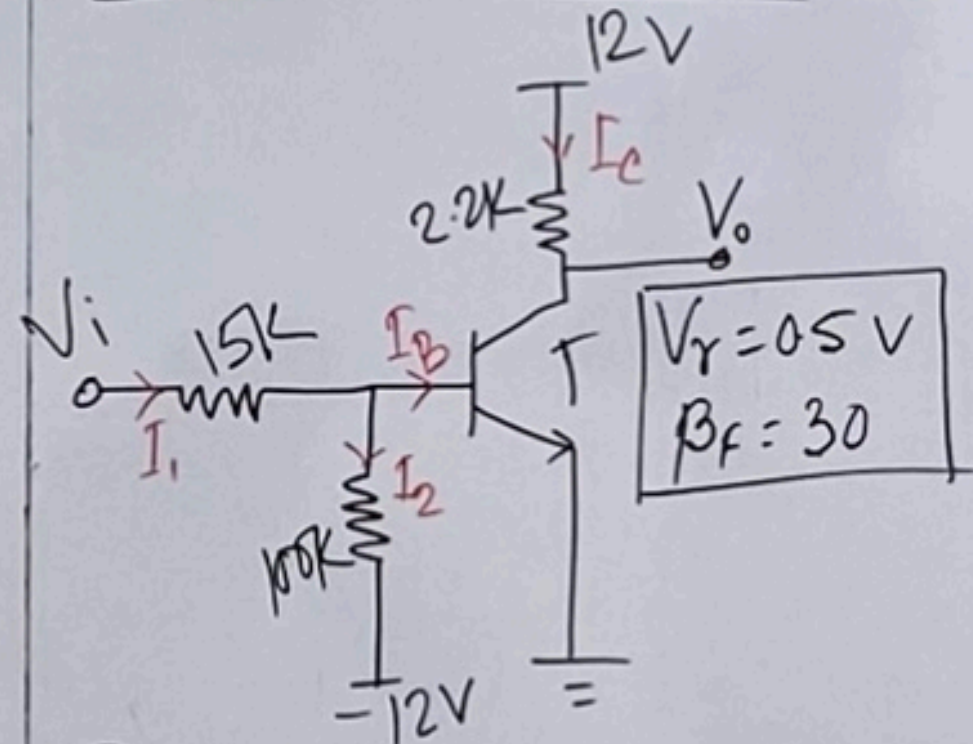


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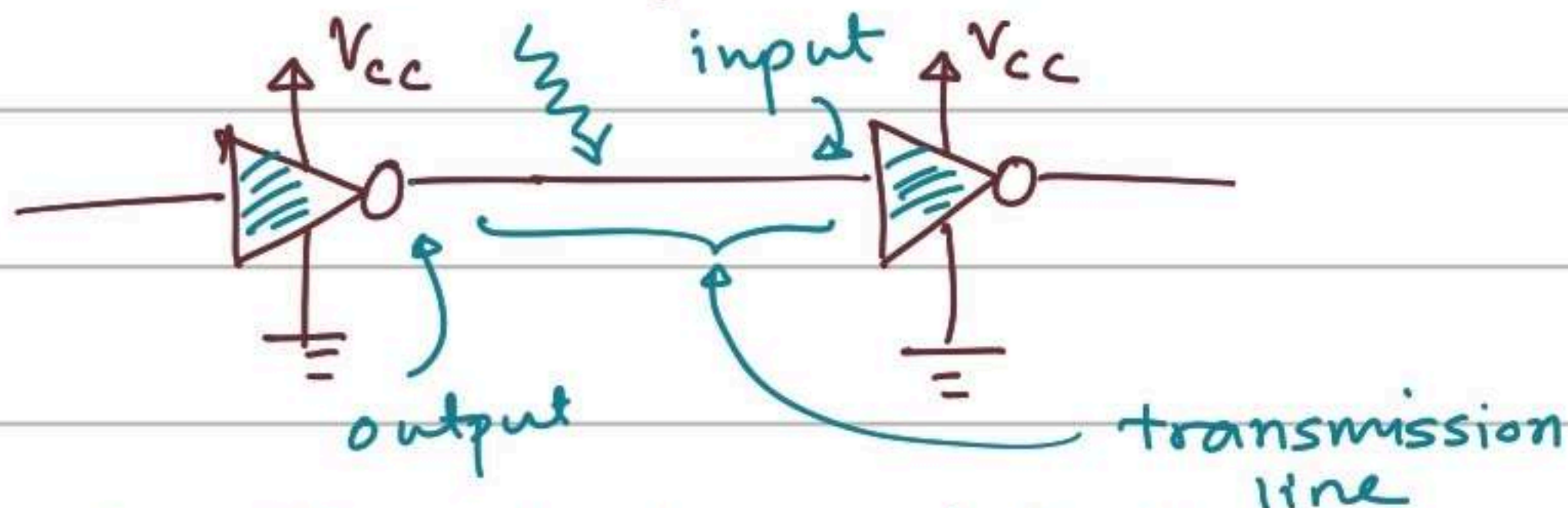
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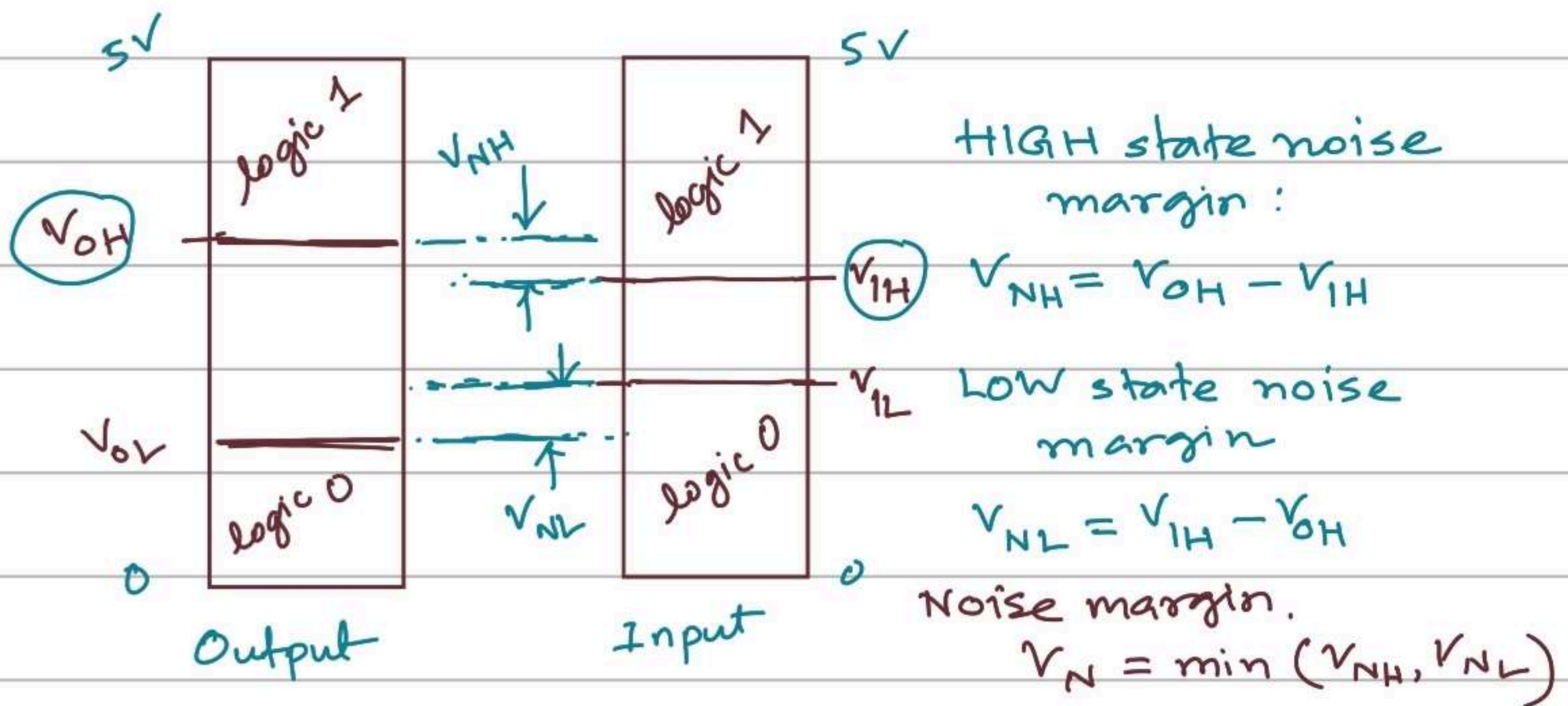
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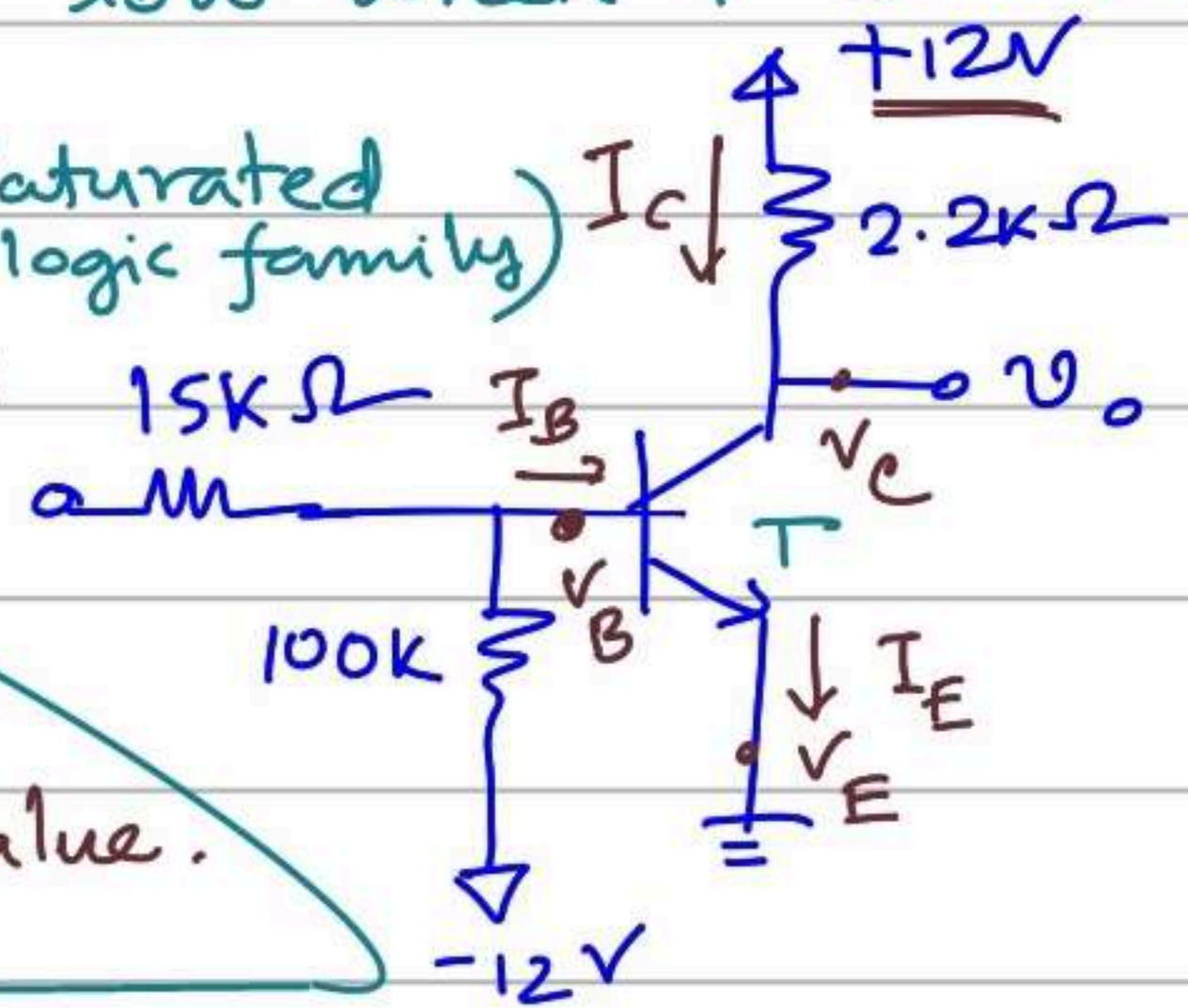
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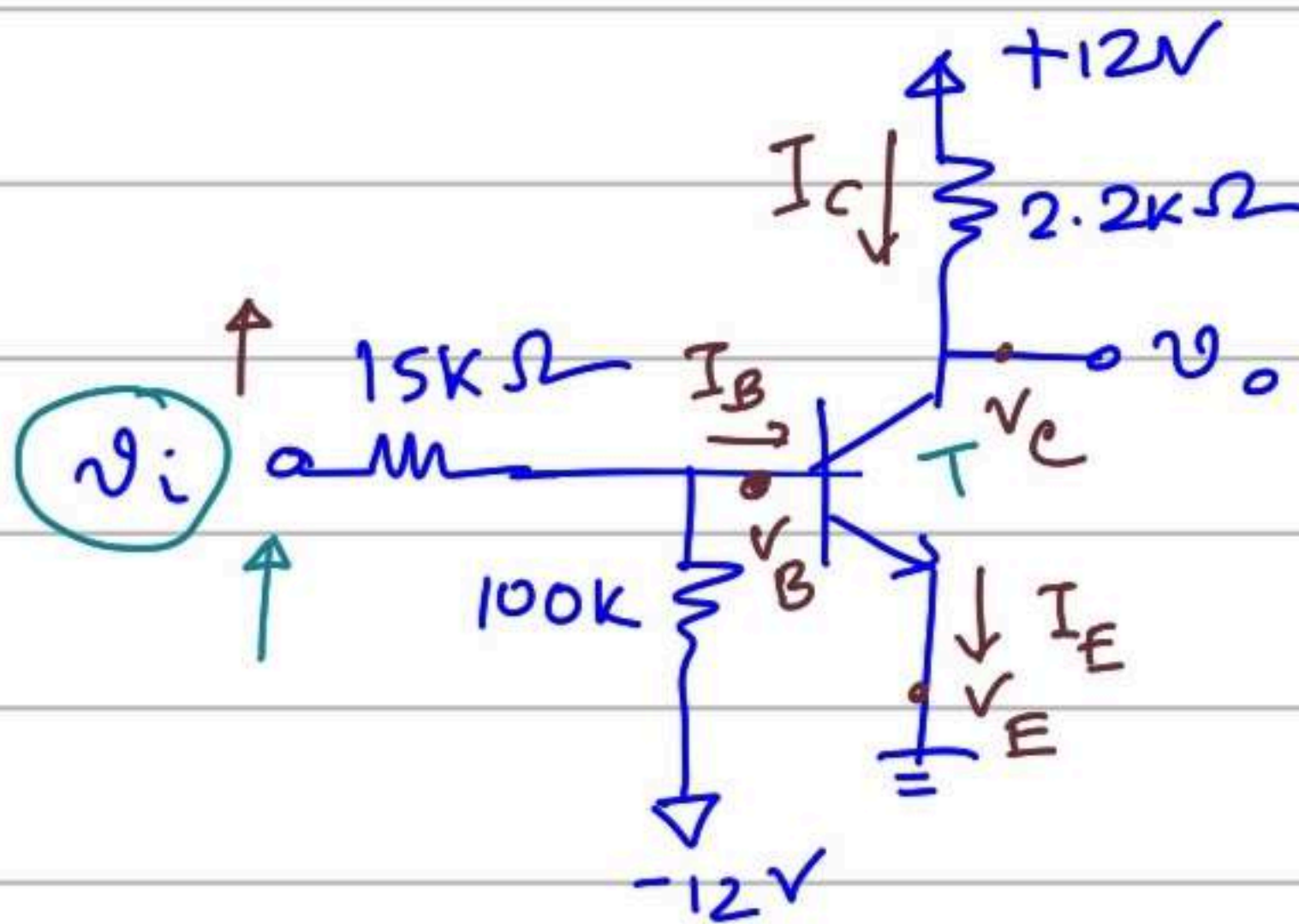
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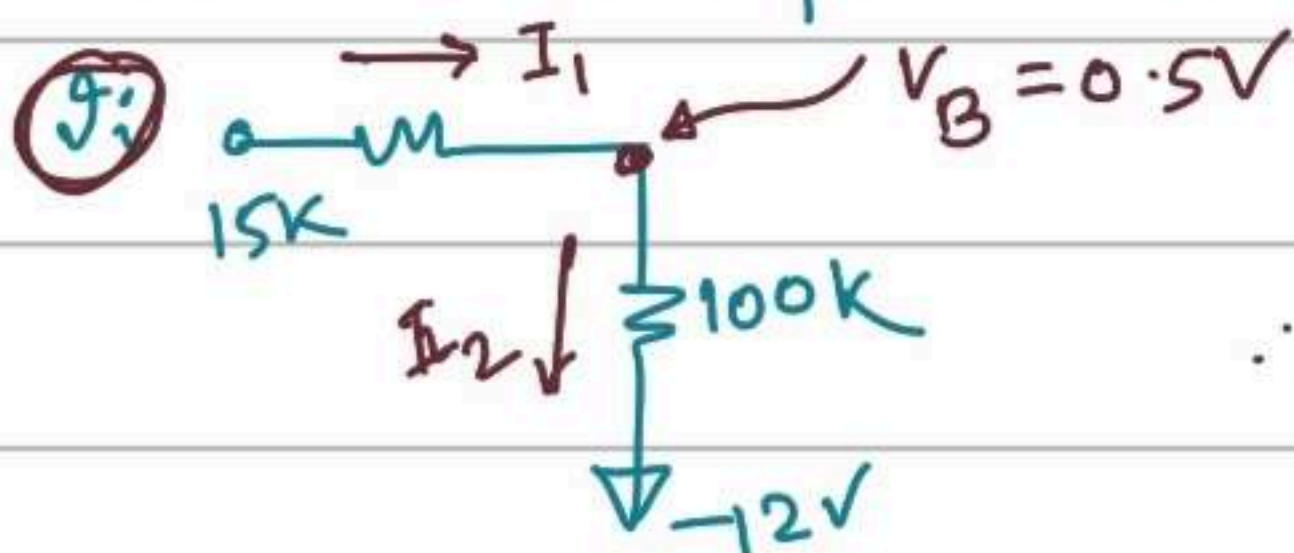
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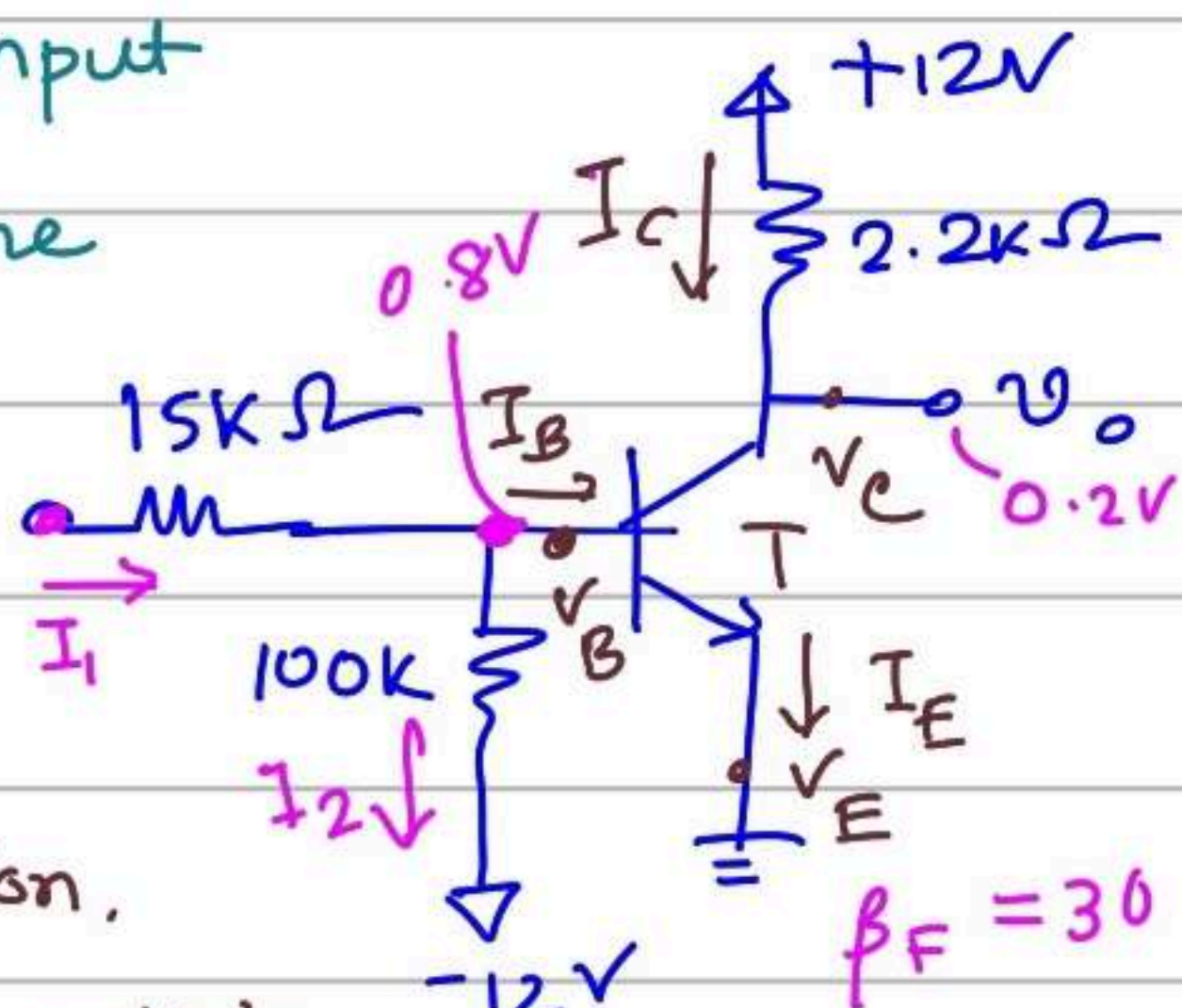
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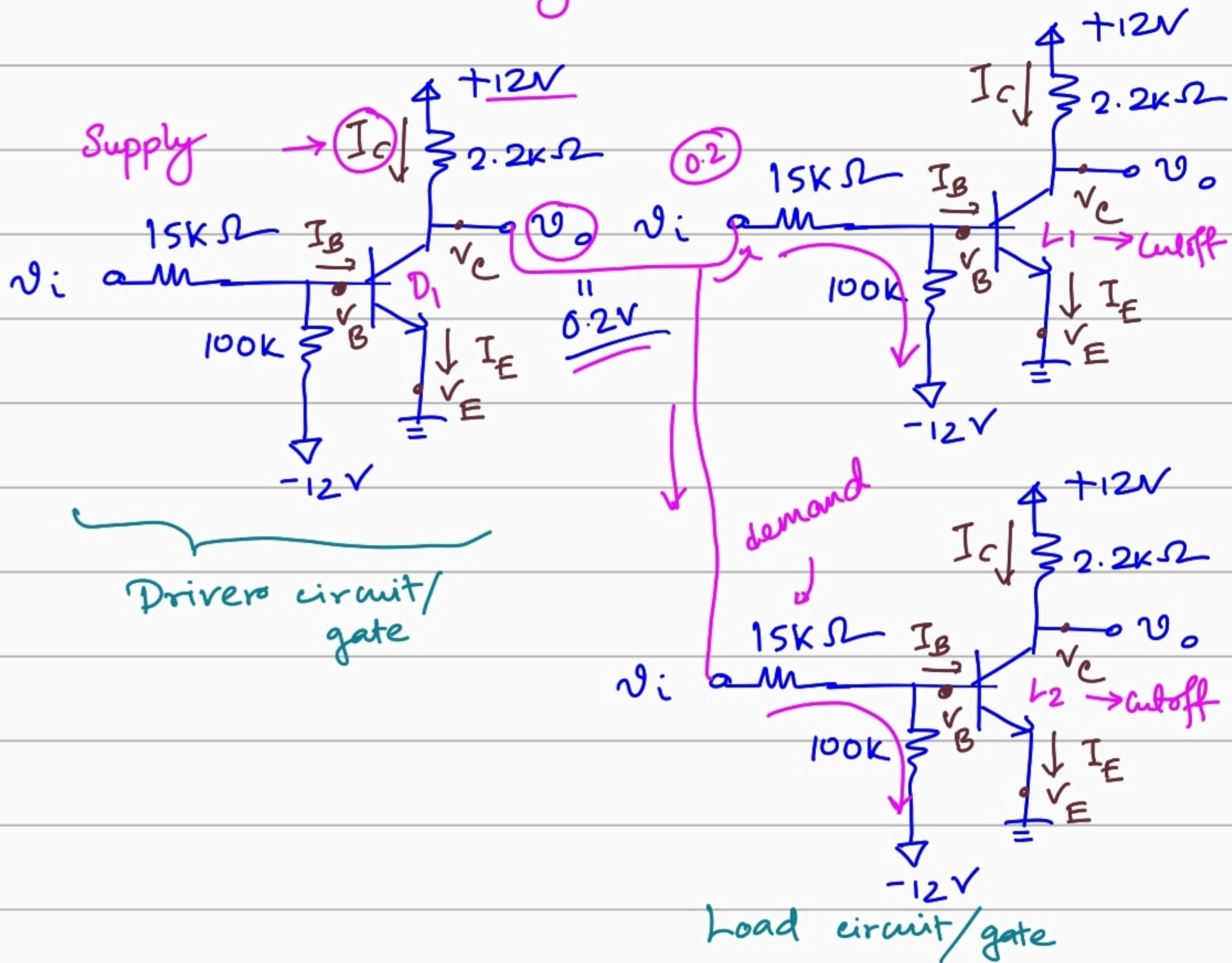
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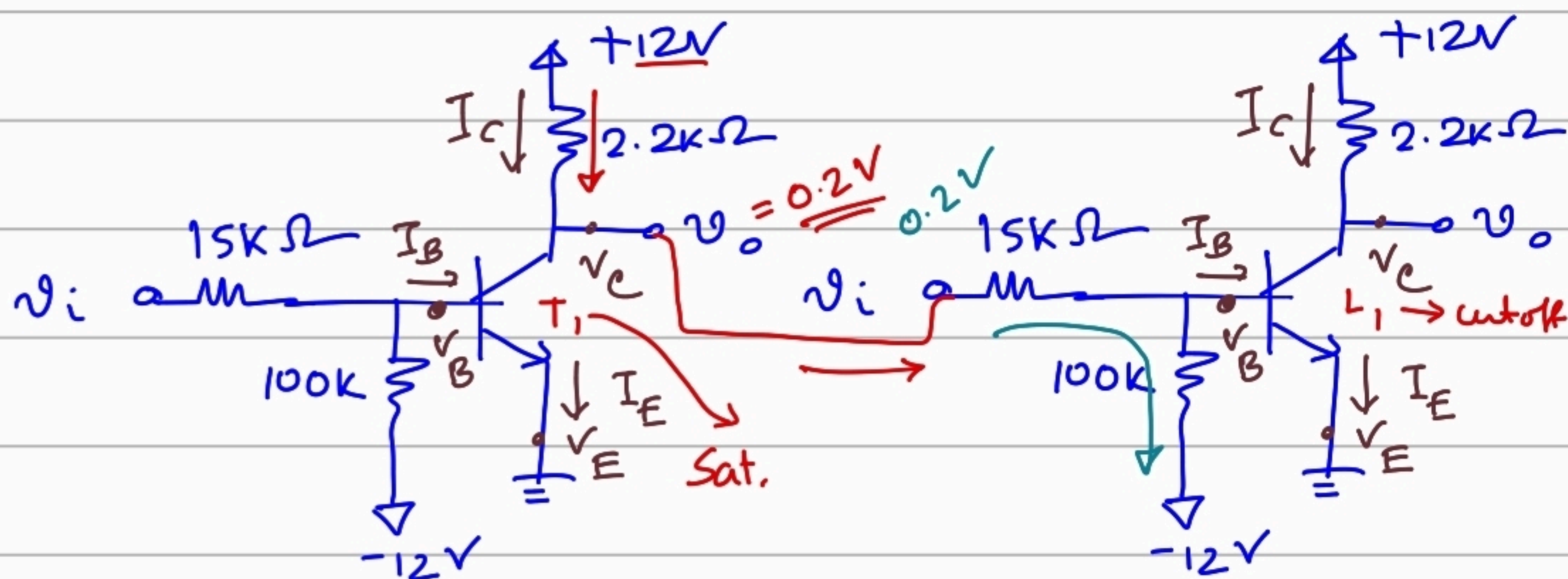


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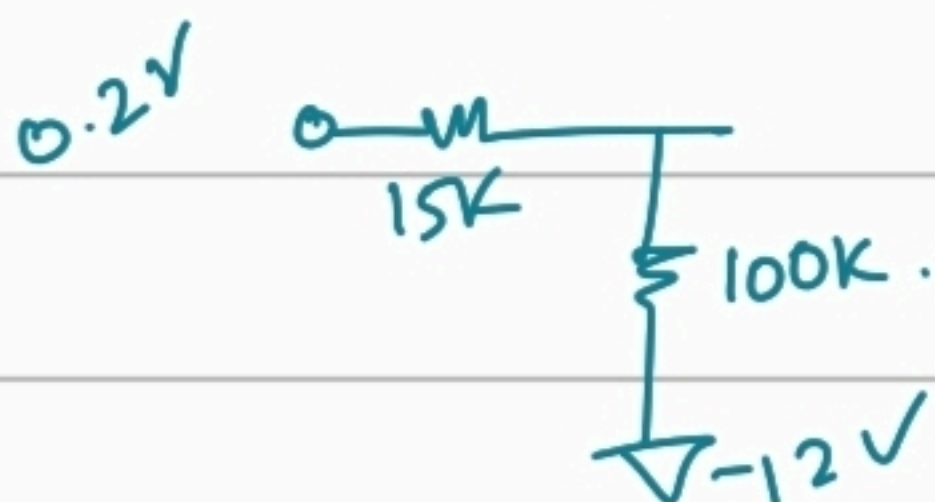


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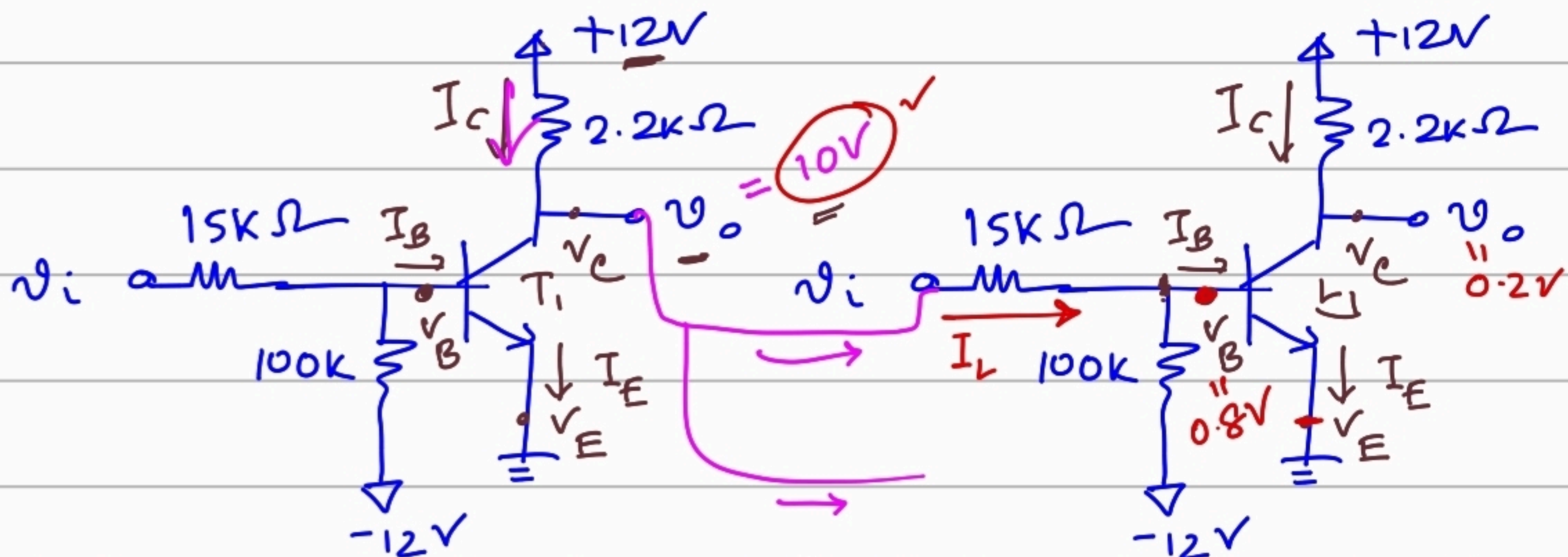
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